

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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INDEX
NUMBER

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CHEMISTRY

9729/02

Paper 2 Structured

11 September 2018

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

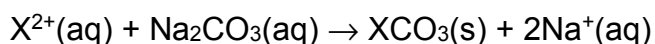
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE			
Paper 1		/ 30	
Paper 2			
1	/ 9	4	/12
2	/ 8	5	/ 10
3	/ 18	6	/ 18
Penalty		sf	unit
Paper 2		/ 75	
Paper 3		/ 80	
Paper 4		/ 55	
Total			

Answer **all** questions in the spaces provided.

- 1 (a)** Water hardness is determined by the concentration of multivalent cations in the water. Common cations found in hard water include Ca^{2+} and Mg^{2+} . Lime softening is a type of water treatment where aqueous Na_2CO_3 is added to remove Ca^{2+} ion as CaCO_3 is precipitated. These precipitates are then removed by sedimentation and filtration, where X^{2+} can be Ca^{2+} and Mg^{2+} .



In the laboratory, 1 cm^3 of 0.50 mol dm^{-3} Na_2CO_3 is added to 10 cm^3 of a raw water sample. What is the minimum concentration of Ca^{2+} in the original raw water sample, which would cause a white precipitate to be observed?

The value of K_{sp} for CaCO_3 is $4.81 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$.

[2]

- (b)** A saturated solution of raw water containing both CaF_2 and $\text{Ca}(\text{OH})_2$ has a pH of 12 at 25°C .

calcium compound	numerical value of K_{sp}
CaF_2	3.2×10^{-11}
$\text{Ca}(\text{OH})_2$	6.4×10^{-6}

- (i)** Calculate the concentration of hydroxide ions in the solution.

[1]

- (ii) Hence, calculate the solubility of CaF_2 in the raw water sample. Show your working clearly.

[2]

- (c) Using the following data, construct an energy cycle to calculate the enthalpy change for the thermal decomposition of calcium carbonate.

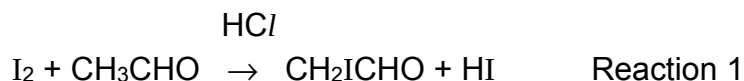


	$\Delta H / \text{kJ mol}^{-1}$
$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	-57.1
$\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{aq})$	-63.7
$\text{H}_2\text{SO}_4(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow \text{CaSO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	-151.5

[4]

[Total: 9]

- 2 The reaction of ethanal with iodine occurs in the presence of acid catalyst, HCl . The reaction is given below.



The table shows the results obtained when the concentrations of I_2 , CH_3CHO and HCl were varied.

Experiment	Initial $[\text{HCl}]$ / mol dm^{-3}	Initial $[\text{I}_2]$ / mol dm^{-3}	Initial $[\text{CH}_3\text{CHO}]$ / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.10	0.20	0.20	1.6×10^{-6}
2	0.10	0.40	0.30	2.4×10^{-6}
3	0.40	0.40	0.20	6.4×10^{-6}
4	0.10	0.20	2.00	1.6×10^{-5}

- (a) (i) Deduce the orders of reaction with respect to I_2 , CH_3CHO and HCl . Hence, determine the rate equation.

[3]

- (ii) The half-life of ethanal in experiment 1 was 60 seconds. Predict the half-life of ethanal in experiments 2 and 3. Explain your answer.

.....

 [2]

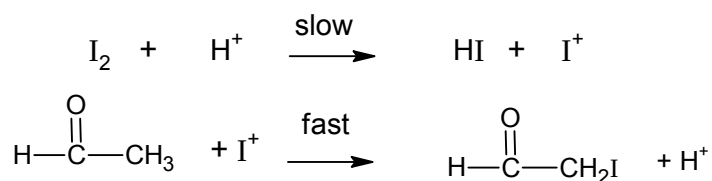
(iii) Sketch the graph of concentration of iodine against time for experiment 4.

Experiment	Initial [HCl] / mol dm ⁻³	Initial [I ₂] / mol dm ⁻³	Initial [CH ₃ CHO] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
4	0.10	0.20	2.00	1.6 × 10 ⁻⁵

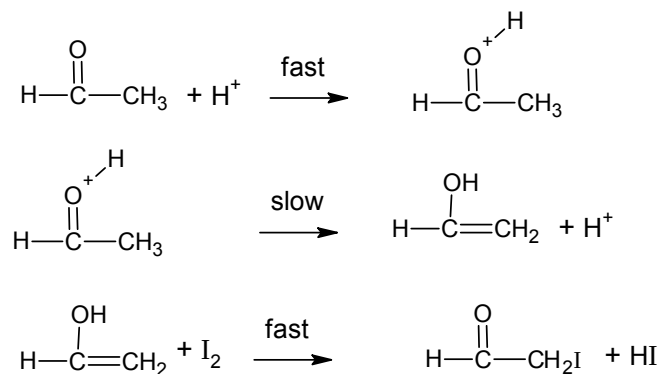
[1]

(b) The following mechanisms **A** and **B** are proposed for the reaction of ethanal, iodine and acid catalyst in Reaction 1.

Mechanism A



Mechanism B



Which of the above mechanisms best agrees with the experimentally determined rate equation in (a)(i)? Explain your answer.

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.....

.....

..... [2]

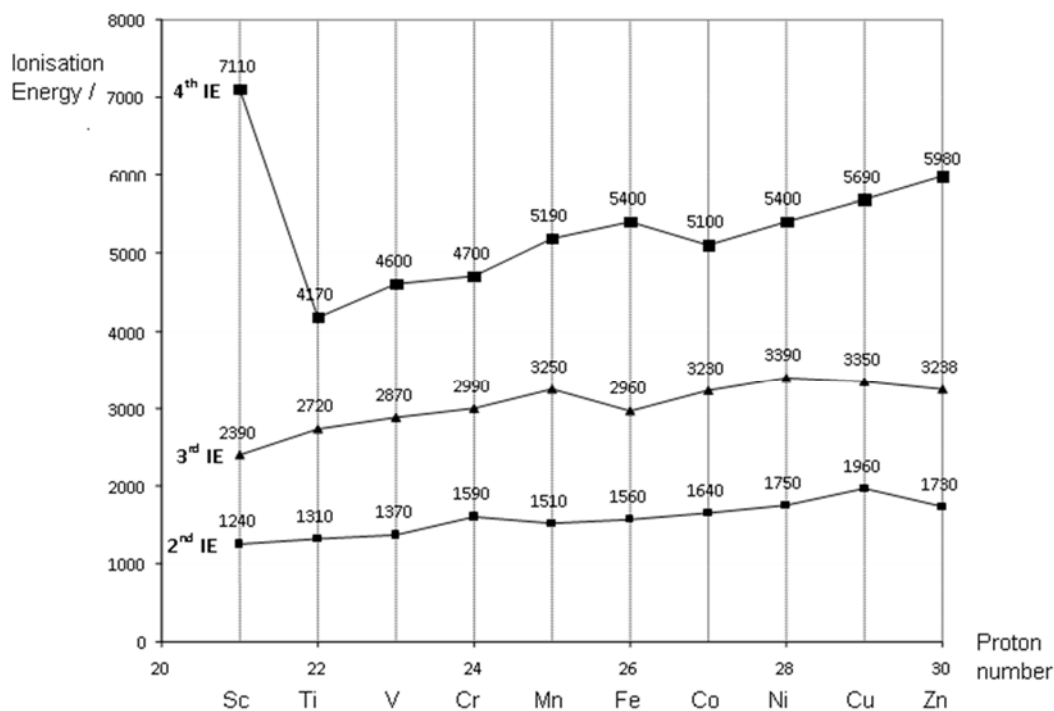
[Total: 8]

- 3 Transition metal is "an element whose atom has a partially filled d sub-shell, or which can give rise to cations with an incompletely filled d sub-shell". Chromium and vanadium, both transition metals, have found a number of industrial uses due to their toughness and resistance to heat and corrosion.

The table below shows some information of a Group 2 metal calcium, vanadium and chromium.

	Vanadium	Chromium	Calcium
Melting point/ °C	1910	1875	843
Atomic radius/ nm	0.135	0.129	0.197
Common Oxidation Number	+2, +3, +4, +5	+2, +3, +6	+2

The graph below shows the second to fourth ionisation energies for the first row d block elements scandium to zinc.



- (a) (i) Although the vanadium atom has more electrons than the calcium atom, the atomic radius of vanadium is smaller than that of calcium. Suggest an explanation for this.

.....

.....

.....

.....[2]

- (ii) Explain why the second ionisation energies of transition elements from Ti to Cu are relatively invariant.

.....

 [2]

- (iii) Explain why the second ionisation energy of chromium is higher than that of manganese.

.....

 [2]

- (b) (i) State the **full** electronic configuration of Cr^{3+} ion.

..... [1]

- (ii) The complex ion $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is green, and it is an octahedral complex. On the Cartesian axes shown in Fig. 2.1, draw **fully-labelled** diagrams of the following:

- One of the d orbitals at the lower energy level in an octahedral complex- Label this diagram 'lower'.
- One of the d orbitals at the upper energy level in an octahedral complex. Label this diagram 'upper'.

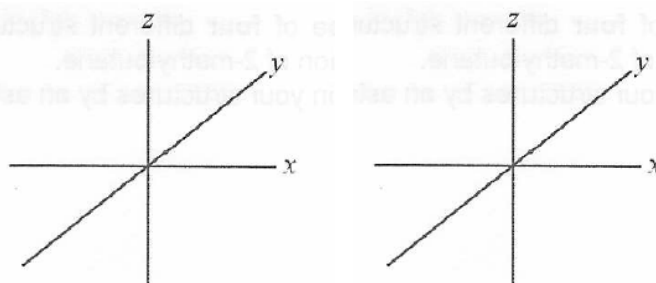


Fig. 3.1

[2]

- (iii) Suggest which of the two compounds chromium(III) oxide, Cr_2O_3 , and chromium(III) carbonate, $\text{Cr}_2(\text{CO}_3)_3$, would have a more exothermic lattice energy. Explain your answer.

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.....
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..... [2]

- (iv) The compound K_3CrO_4 is a green solid. When mixed with dilute H_2SO_4 , it undergoes disproportionation to yield $\text{Cr}^{3+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ ions.

By writing appropriate half-equations, write an overall balanced equation for the disproportionation of CrO_4^{3-} in acid.

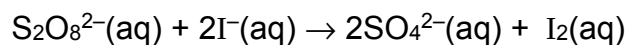
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..... [2]

- (v) Purification of chromium can be achieved by electro-refining chromium from an impure chromium anode onto a pure chromium cathode in an electrolytic cell. How many hours will it take to plate 130 g of chromium onto the cathode if the current passed through the cell is held constant at 34.0 A?

Assume the chromium in the electrolytic solution is present as Cr^{3+} .

[2]

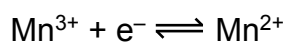
- (c) The reaction between $\text{S}_2\text{O}_8^{2-}$ ions and I^- ions is very slow:



If a small amount of aqueous manganese(II) ions, Mn^{2+} , is added to the mixture, the rate of reaction increases rapidly.

State the property, typical of transition metals, which allow manganese (II) ions to behave as a homogeneous catalyst in this reaction.

By considering the data given below, write relevant chemical equations to support your answer.



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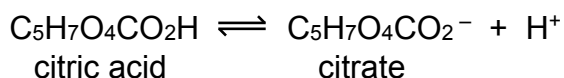
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[3]
[Total:18]

- 4 (a) Citric acid is commonly found in citrus fruits such as lemons. Mixtures of citric acid and its salt, sodium citrate, are often used as “acidity regulators”. These are food additives that have a buffering action on the pH of foodstuffs.

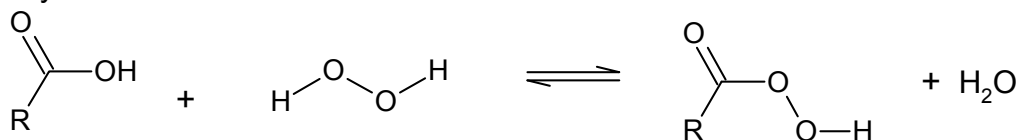


$$K_a \text{ of citric acid} = 7.4 \times 10^{-4} \text{ mol dm}^{-3}$$

A typical citric acid / sodium citrate buffer mixture is prepared by mixing $0.100 \text{ mol dm}^{-3}$ citric acid and $0.200 \text{ mol dm}^{-3}$ sodium citrate in a volume ratio 2:3 respectively. Calculate the pH of the buffer solution.

[2]

- (b) The reaction between carboxylic acids and hydrogen peroxides produce peroxyacids.



The pK_a values of four compounds are listed in the table below.

name	formula	pK_a
Water	H_2O	14.0
Hydrogen peroxide	H_2O_2	11.7
Methanoic acid	HCO_2H	3.7
Peroxymethanoic acid	HCO_3H	7.1

- (i) Suggest an explanation for why
The pK_a of H_2O_2 is less than that for water

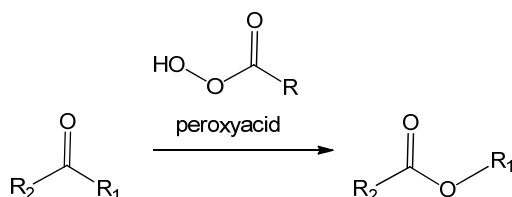
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..... [1]

- (ii) the pKa of HCO_3H is more than that for HCO_2H

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 [2]

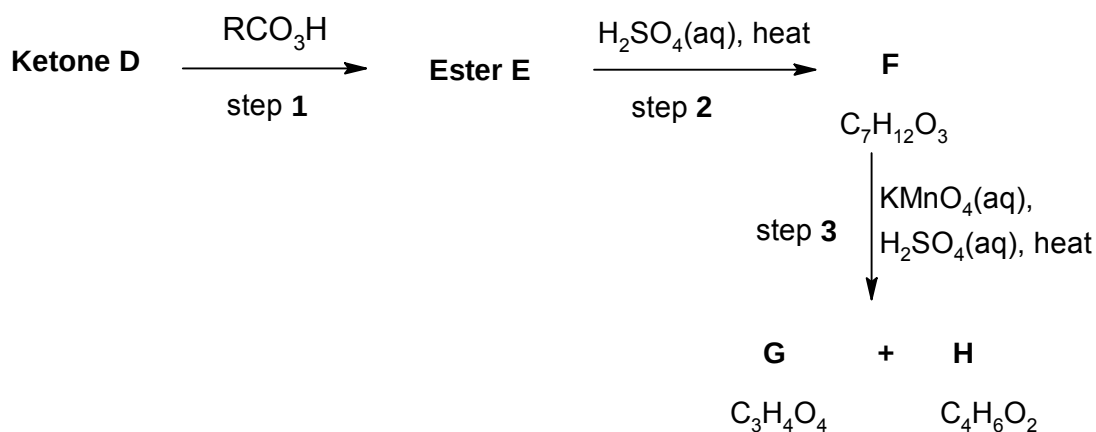
- (c) The Baeyer-Villiger oxidation is an organic reaction in which peroxyacids react with ketones to give esters, as shown in the equation below.



In the following scheme, ketone **D** ($\text{C}_7\text{H}_{10}\text{O}$) undergoes Baeyer-Villiger oxidation in step 1 to form ester **E**. Ketone **D** also decolourises orange aqueous Br_2 .

F is the **only** product formed, from treating ester **E** with hot $\text{H}_2\text{SO}_4(\text{aq})$.

Hot acidified KMnO_4 oxidises several classes of organic compounds to ketones, carboxylic acids and carbon dioxide. By this means, the structures of compounds can be determined. The scheme below shows the oxidation products of **F**.



- (i) State the type of reaction Ester **E** undergoes with hot $\text{H}_2\text{SO}_4(\text{aq})$ in step 2.

Type of reaction: [1]

- (ii) 1 mole **H** reacted with alkaline aqueous iodine to form 2 moles of CHI_3 yellow ppt. Suggest the structure of **H**.
Give the product of the reaction of **H** with alkaline aqueous iodine.
Structure of **H**:

Product:

[2]

- (iii) **G** ($\text{C}_3\text{H}_4\text{O}_4$) was found to react with $\text{Na}_2\text{CO}_3(\text{aq})$. Suggest the structure of **G**.

[1]

- (iv) From your answer in (ii) and (iii), deduce and suggest the structure of **F**.

[1]

- (v) Hence, suggest the structures of ketone **D** and ester **E**.

[2]

[Total: 12]

- 5 (a) (i) Oleic acid, $\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{CO}_2\text{H}$ is a fatty acid found in oils and fats, which has C=C double bond in the *cis* configuration.

The *trans* isomer of oleic acid is known as *elaidic acid*, which is found in partially hydrogenated vegetable oils.

Draw the skeletal formula for the ***trans*** isomer of oleic acid, $\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{CO}_2\text{H}$, which is also known as *elaidic acid*.

[1]

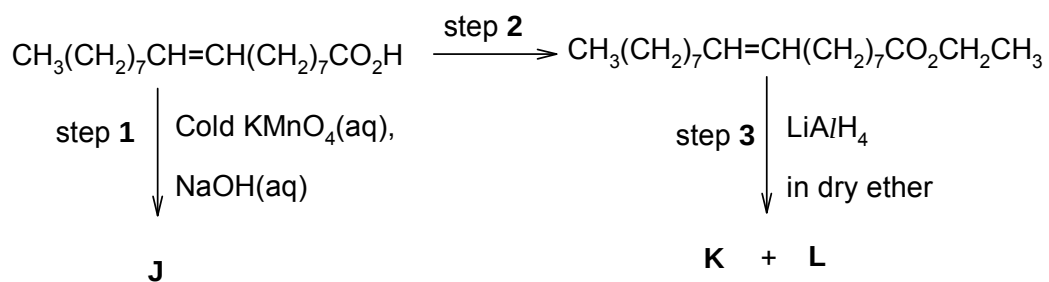
- (ii) Suggest a brief explanation for the differences in melting points between oleic acid and its *trans* isomer, elaidic acid.

Compound	<i>Mr</i>	Melting point / °C
oleic acid (<i>cis</i> -isomer)	282	4.0
elaidic acid (<i>trans</i> -isomer)	282	45.0

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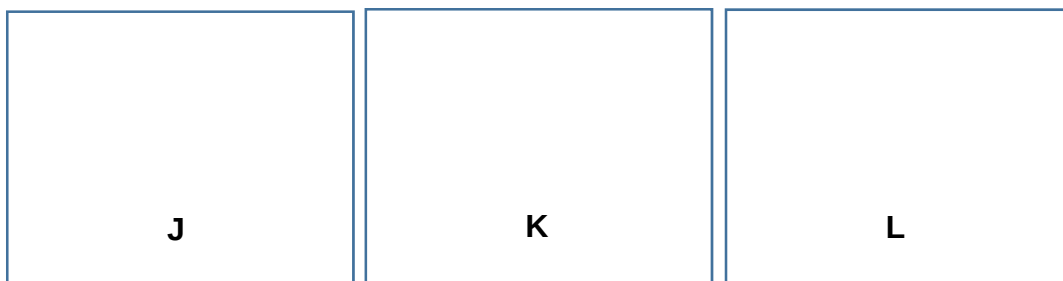
 [1]

(iii) Below outlines a reaction scheme involving oleic acid.



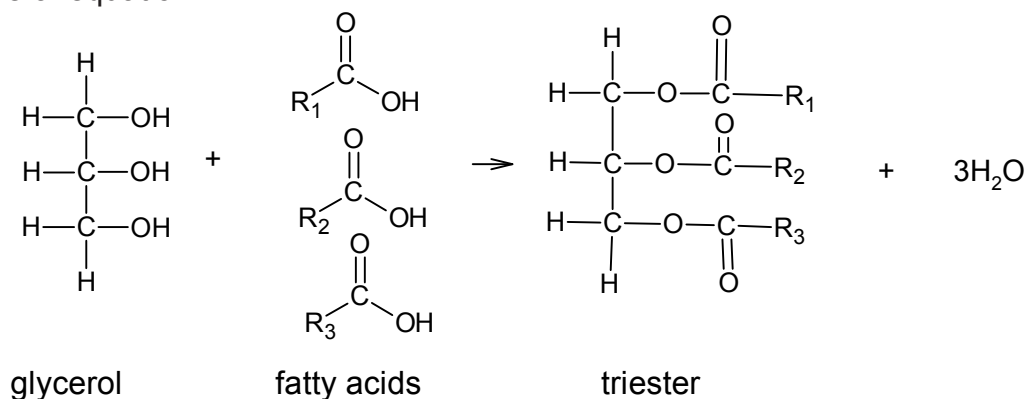
Give the reagents and conditions for step 2, and draw the structures of compounds **J**, **K** and **L** in the boxes provided below.

Step 2:



[4]

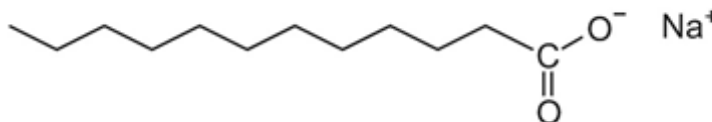
- (b) Avocado oil consists mainly of triesters formed from glycerol (propane-1,2,3-triol) and long-chain carboxylic acids (fatty acids) as shown by the general equation.



The major component of avocado oil is triolein, the triester formed from glycerol and three units of oleic acid (fatty acid).

- (i) A student tried to flush his cup with water to wash away the avocado oil, but was not successful. The student then tried using soap to wash away the avocado oil in the cup, and was successful.

The following compound, sodium stearate is a typical soap for washing.



By reference to the type of intermolecular forces formed between the soap, avocado oil and water, explain why sodium stearate is able to wash away the avocado oil in the cup.

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.....

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.....[2]

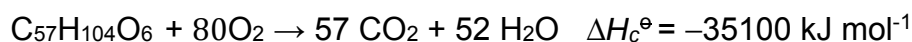
- (ii) Below shows the nutrition content of fats, carbohydrates and protein present in one serving of avocado milkshake.

	Mass present in 1 serving of milkshake	Fuel value/ kJ g ⁻¹
Total Fat	30 g	
Saturated fat (tristearin)	6 g	42.4
Unsaturated fat (triolein)	24 g	x
Trans fat	0 g	-
Carbohydrates	18g	17
Protein	2g	17
Dietary Fiber	16g	8
Potassium	850 mg	-

Fats constitute the bulk of the milkshake, and are metabolised into carbon dioxide and water when subjected to combustion in a bomb calorimeter. There are two types of fats in the milkshake, saturated versus unsaturated fat.

You may assume that the unsaturated fat and saturated fat in the milkshake are triolein and tristearin respectively.

The combustion of triolein, C₅₇H₁₀₄O₆, the unsaturated fat is as follows:



The **fuel value** of a substance is defined as the heat energy released, when **one gram** of the substance undergoes combustion.

Calculate x, the fuel value of triolein, the unsaturated fat, and hence deduce if triolein or tristearin is a better source of energy (*M_r* of triolein = 884).

[2]
[Total: 10]

6 Aluminium oxide is frequently used in organic reactions and is also a common component in cosmetics. It is primarily used in the production of aluminium.

(a) Aluminium oxide reacts with aqueous sodium hydroxide to give a homogeneous solution. Similarly, aluminium reacts with aqueous sodium hydroxide to give the same homogeneous solution and a gas that pops with a lighted splint.

(i) Write a balanced equation for the reaction of a piece of aluminium foil when it is placed into aqueous sodium hydroxide.

..... [1]

(ii) In reaction (a)(i), a white solid layer was first seen on the aluminium foil before the homogeneous solution was formed. Identify the solid and write an equation to explain how the homogeneous solution was formed from the solid.

Solid:

.....

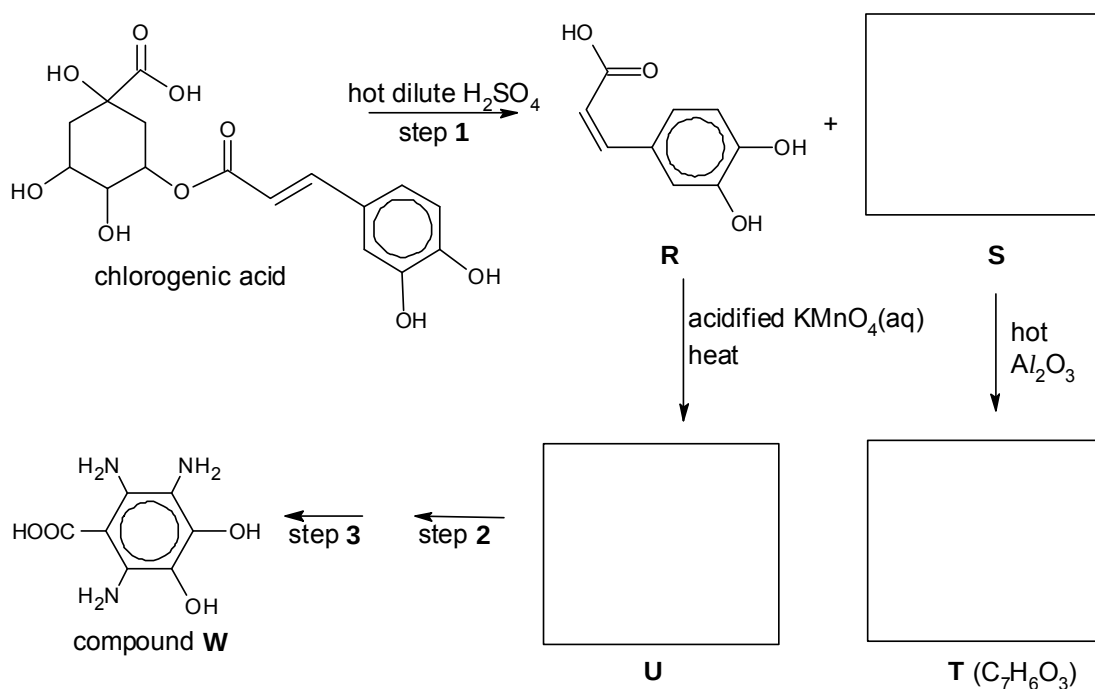
.....

..... [2]

(b) Polyphenols are often reducing agents known as antioxidants.

An example of a polyphenol is chlorogenic acid.

The structure of chlorogenic acid and its reaction scheme is shown below. In the reaction scheme, compound **T**, $C_7H_6O_3$, is formed from compound **S** and heated Al_2O_3 . Compound **T** reacts with neutral aqueous iron(III) chloride to form a violet solution.



(i) Draw the structures of organic compounds **S**, **T** and **U** in the boxes provided in the scheme above. [3]

(ii) Compound **U** can be converted to compound **W** in two steps. Suggest reagents and conditions for steps 2 and 3.

Step 2: [1]

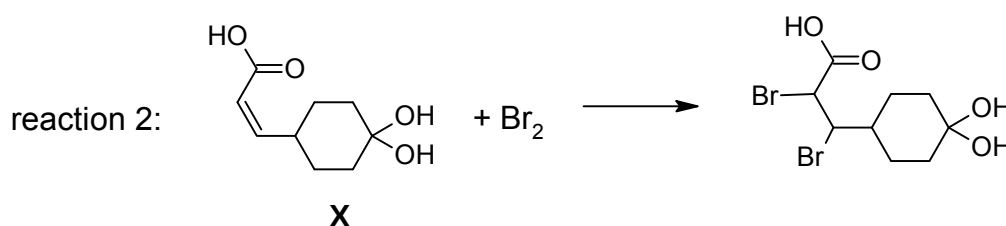
Step 3: [1]

- (iii) If hot dilute sodium hydroxide was used to react with chlorogenic acid, state the number of moles of NaOH that would be required to react with one mole of chlorogenic acid.

Number of mol of NaOH:

[1]

- (iv) Compound **X** shown below is an analog of compound **R** in the scheme. When compound **X** is added to bromine in CCl_4 in the dark, there is a rapid reaction 2, that decolourises the bromine.

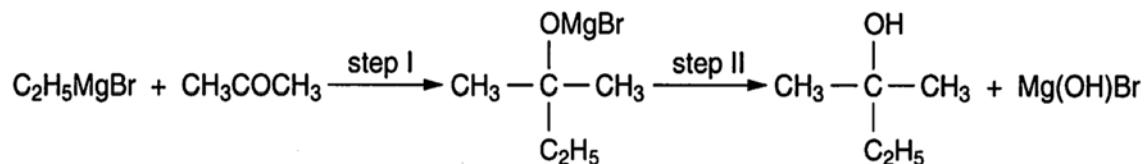


The product of this reaction 2, exists as a mixture of four stereoisomers. State the type of isomerism exhibited and draw all the stereoisomers.

[3]

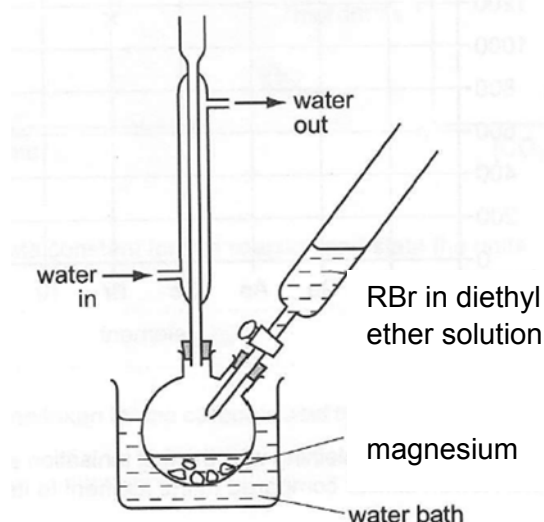
- (c) In a Grignard reaction, the Grignard reagent, RMgBr (alkyl or aryl magnesium halide), may be added to a carbonyl compound for the formation of an alcohol.

A typical example of the use of a Grignard reagent is the two-step reaction of $\text{C}_2\text{H}_5\text{MgBr}$ with propanone, CH_3COCH_3 , to form 2-methylbutan-2-ol.



- (i) Draw the structural formula of the Grignard reagent that will react with propanone to form 2-methylhexan-2-ol in a similar two-step reaction. [1]

- (ii) To prepare 2-methylhexan-2-ol, the following set-up and procedure was used.



Preparation of the Grignard reagent:

- Several small pieces of Mg was placed in a round-bottomed flask.
- 9.6 g of an appropriate bromoalkane, RBr, was dissolved in anhydrous diethyl ether. It is important to exclude water as Grignard reagent undergoes acid-base reaction with water rapidly to form an alkane, destroying the Grignard reagent.
- To initiate the reaction, add just enough RBr/ ether solution from the dropping funnel to cover the magnesium. Start stirring with a magnetic stir bar and use a water bath to heat the reaction flask as the RBr/ ether solution is slowly added dropwise into the metal.
- The mixture was heated under reflux for about 15 minutes and cooled in an ice-bath.

Reaction of Grignard reagent with propanone:

5. Add 1.2 g of anhydrous propanone in the dropping funnel and add propanone dropwise to the Grignard reagent.
6. Allow the reaction mixture to stand at room temperature for 30 minutes.
7. The mixture was then slowly poured with vigorous stirring into an ice-cold substance **Z** in a beaker. Almost immediately, a solid precipitate is seen.
8. Filter the mixture and collect the ethereal solution in a clean and dry conical flask.
9. Add anhydrous magnesium sulfate to the ethereal solution, cork the flask tightly and leave the mixture to stand for a while.
10. Filter and collect the filtrate in a round-bottom flask which is then placed on a rotary dryer to evaporate all the solvent diethyl ether away.

I. Assuming that 1.9 g of 2-methylhexan-2-ol was prepared using the procedure, calculate the percentage yield for this synthesis. [2]

II. During the preparation of the Grignard reagent, anhydrous diethyl ether was used as the solvent in step 2, as the Grignard reagent can be destroyed in the presence of water.

Write an equation to show the reaction of the Grignard reagent with water.

..... [1]

III. Identify substance **Z** in step 7, in the beaker.

..... [1]

IV. Suggest what is removed when anhydrous magnesium sulfate is added to the ethereal solution in step 9.

..... [1]

[Total: 18]

End of paper